Botany Labs

Botany Labs

GRADE(S): 9-12

About the Course

Note that labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

This topic is included in the following course(s): Science: Botany

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Botany Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Botany				
Botany Lessons Nature Walks & Scouting: Grades 9-12	Botany Lessons	Botany Lessons	Botany Lessons Nature Notebook: Grades 9-12	Botany Lessons Botany Labs

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies
∞ View Supply List Details

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Slides and Coverslips

Quick Links

(No Quick Links Assigned)

Click THIS text or scan the QR code for links.

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How To Approach Labs

Introduce

● Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.

● Relevant concept(s) are introduced in the text.

● Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.

Lab Procedure

● Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.

● Record all data and observations in the lab notebook.

Analysis & Conclusions

● After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.

● If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

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Click THIS text or scan the QR code for links.

Term 1

WEEK 1 ⬚ **60m Botany Labs - Lesson 1**

▯ Materials: Elementary Studies in Plant Life

Prep: Read Student Tip.

→ OBSERVE
Notice a variety of plants this week, doing the practical work from Ch.1. Notice root and shoot, and how the leaves and branches are attached in the different cases. Notice plants in which flowers are fading, and fruit is ripening in their place. Some activities may be delayed depending on the time of year and your geographic location. Do what you can to gain the most from the chapter. If the plants or trees mentioned are unavailable locally, look up the item online. Also, you may need to visit local nurseries, botanical gardens, and neighbors' yards and gardens to find specimens to examine.

★ STUDENT TIP
Labs and prompts for your field work are included in your lesson plans, but it is important to establish a personal habit of outdoor walks as a part of life!

WEEK 2 ⬚ **60m Botany Labs - Lesson 2**

▯ Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform the practical work from Ch.2. Record your work in your notebook.

WEEK 3 ⬚ **60m Botany Labs - Lesson 3**

▯ Materials: Link, household items indicated, microscope, and glass slides

Prep: Read Student Tip.

→ LAB DAY
Use the procedure at the link to evaluate microbial life in soil from at least 3 locations. The locations may be at the same site, but they should be some distance apart. Soil that is even 10-20 yards apart may be quite different. It is okay to be unsure of what you are viewing under the microscope.

→ VIEW
∞ Link: Soil Sampling Procedure

★ STUDENT TIP
Plan to go to another type of habitat next week. If you collected soil from a neighborhood this week, perhaps go to a forest or a pond next week.

WEEK 4 ⬚ **60m Botany Labs - Lesson 4**

▯ Materials: Link, household items indicated, microscope, and glass slides

→ LAB DAY
This week, go to a different habitat. Use the procedure at the link to evaluate microbial life in soil from at least 3 locations.

→ VIEW
∞ Link: Soil Sampling Procedure

WEEK 5 ⬚ **60m Botany Labs - Lesson 5**

▯ Materials: Elementary Studies in Plant Life, pea/bean seeds, and potting materials

→ LAB DAY
Perform the practical work from Ch.3. Record your work in your notebook. Also, take a few minutes to start some pea or bean seeds for Week 10. Be sure to keep them watered!

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Term 1

WEEK 6

- [ ] **60m Botany Labs - Lesson 6**

> Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform a selection of the practical work from Ch.17. Record your work in your notebook.

WEEK 7

- [ ] **60m Botany Labs - Lesson 7**

> Materials: Link, at least one lily, and a dissecting kit (supplies noted in lab)

→ DISSECTION
Procure a lily flower. You might have to purchase one at a florist or local grocery store. Read the guide linked below, and then dissect your flower. Sketch and label all the parts using the dissection guide. Have a magnifying lens handy to take a close-up look.

→ VIEW
∞ Link: Dissection Guide

WEEK 8

- [ ] **60m Botany Labs - Lesson 8**

> Materials: Elementary Studies in Plant Life, pea/bean seeds, and potting materials

→ LAB DAY
Perform the practical work from Ch.5. Record your work in your notebook. Also, take a few minutes to start some pea or bean seeds for Week 10. Be sure to keep them watered!

WEEK 9

- [ ] **60m Botany Labs - Lesson 9**

> Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform the practical work from Ch.6. Record your work in your notebook.

WEEK 10

- [ ] **60m Botany Labs - Lesson 10**

> Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform the practical work from Ch.7. Record your work in your notebook.

WEEK 11

- [ ] **60m Botany Labs - Lesson 11**

> Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform the practical work from Ch.8. Record your work in your notebook.

WEEK 12

- [ ] **60m Botany Labs - Lesson 12**

Exam Week

→ EXAMS
Answer question(s) related to the course.

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Term 2

WEEK 1 ⬚ **60m Botany Labs - Lesson 13**

Materials: Elementary Studies in Plant Life

➔ LAB DAY
Perform the practical work from Ch.9. Record your work in your notebook.

WEEK 2 ⬚ **60m Botany Labs - Lesson 14**

Materials: Link

➔ PLAN
Explore citizen science projects, such as Tree Snap, The Christmas Bird Count, BudBurst, and more. Choose one to do during the term that is related to trees or the creatures that depend on trees. Mark your calendar and obtain any supplies needed for the project. If you have another lab that week, you may shift your lab schedule to accommodate the project.

➔ VIEW
∞ Link: SciStarter

WEEK 3 ⬚ **60m Botany Labs - Lesson 15**

Materials: Elementary Studies in Plant Life

➔ LAB DAY
Perform the practical work from Ch.15.

WEEK 4 ⬚ **60m Botany Labs - Lesson 16**

Materials: Sibley Guide to Trees

➔ READ, NARRATE, & DISCUSS
"Twigs" and "Buds" p.xxxv-xxxvii

➔ PRACTICE
Identify the features described among the twigs you have collected.

WEEK 5 ⬚ **60m Botany Labs - Lesson 17**

Materials: Sibley Guide to Trees

➔ READ, NARRATE, & DISCUSS
"Bark" p.xxxvii-xxxviii

➔ PRACTICE
Identify the features described among various trees.

WEEK 6 ⬚ **60m Botany Labs - Lesson 18**
Catch-Up Day

➔ PRACTICE
Catch-up day to accommodate citizen science.

WEEK 7 ⬚ **60m Botany Labs - Lesson 19**
Catch-Up Day

➔ PRACTICE
Catch-up day to accommodate citizen science. Or, if you are ready, move on to next week's project!

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Term 2

WEEK 8 ⬚ **60m Botany Labs - Lesson 20**

Materials: Seed collection or ethnobotany project, student choice

➜ PRACTICE
For the last several weeks of the term, you will work on 1 of 2 projects:
1. Create a "seed bank" to preserve and present your collection to your teacher; or
2. Choose one of your local plants, and research how the plant was used culturally by a group of people. Present your findings to your teacher. The video link below contains a great example of ethnobotany.

➜ VIEW
∞ Video Link: Ethnobotany (6:09)

Decide today which project you will do and what form it will take. A seed bank could be a collection of paintings, models/sculptures, poems, musical motifs, etc. Ethnobotany research could be presented as a brochure designed on Canva, a slide/poster presentation, a fireside tale, etc. Once you are ready, begin working!

WEEK 9 ⬚ **60m Botany Labs - Lesson 21**

Materials: Seed collection or ethnobotany project, student choice

➜ PRACTICE
Work on your project.

WEEK 10 ⬚ **60m Botany Labs - Lesson 22**

Materials: Seed collection or ethnobotany project, student choice

➜ PRACTICE
Work on your project.

WEEK 11 ⬚ **60m Botany Labs - Lesson 23**

Materials: Seed collection or ethnobotany project, student choice

➜ PRACTICE
Work on your project.

WEEK 12 ⬚ **60m Botany Labs - Lesson 24**
Exam Week

➜ EXAMS
Answer question(s) related to the course.

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Term 3

WEEK 1 ⬚ **60m Botany Labs - Lesson 25**

Materials: Elementary Studies in Plant Life or link

→ LAB DAY
Perform selected practical work from Ch.11-12. Record your work in your notebook.

OR

→ PLAN
Choose another botany-related citizen science project for spring.

→ VIEW
∞ Link: SciStarter

WEEK 2 ⬚ **60m Botany Labs - Lesson 26**

Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform selected practical work from Ch.11-12. Record your work in your notebook.

WEEK 3 ⬚ **60m Botany Labs - Lesson 27**

Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform the practical work from Ch.13. Record your work in your notebook.

OR

→ PRACTICE
Catch-up day to accommodate citizen science.

WEEK 4 ⬚ **60m Botany Labs - Lesson 28**

Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform the practical work from Ch.14. If you would like to try Question 1, you may look for green ash, white ash, or walnut instead of horse chestnut. Record your work in your notebook.

WEEK 5 ⬚ **60m Botany Labs - Lesson 29**

Materials: Botany in a Day

→ OBSERVE
As spring arrives, try to find an example of each family described in Botany in a Day. Record your observations.

WEEK 6 ⬚ **60m Botany Labs - Lesson 30**

Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform the practical work from Ch.19. Record your work in your notebook.

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Term 3

WEEK 7

⬚ **60m Botany Labs - Lesson 31**

Materials: Elementary Studies in Plant Life

➔ LAB DAY
Perform selected practical work from Ch.20. Record your work in your notebook.

WEEK 8

⬚ **60m Botany Labs - Lesson 32**

Materials: Link, household items indicated, microscope, and glass slides

➔ LAB DAY
Return to one of your soil sampling locations from Term 1. Use the procedure at the link to evaluate microbial life in soil from at least 3 locations and compare to your observations from Term 1.

➔ VIEW
∞ Link: Soil Sampling Procedure

WEEK 9

⬚ **60m Botany Labs - Lesson 33**

Materials: Elementary Studies in Plant Life

➔ LAB DAY
Perform selected practical work from Ch.21. Record your work in your notebook.

WEEK 10

⬚ **60m Botany Labs - Lesson 34**

Materials: Botany in a Day

➔ OBSERVE
As spring arrives, try to find an example of each family described in Botany in a Day. Record your observations.

WEEK 11

⬚ **60m Botany Labs - Lesson 35**
Catch-Up Day

➔ CATCH UP
Use this time to catch up in this or any other course.

WEEK 12

⬚ **60m Botany Labs - Lesson 36**
Exam Week

➔ EXAMS
Answer question(s) related to the course.